Cross-Sensor SAR Image Target Detection Based on Dynamic Feature Discrimination and Center-Aware Calibration

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***Abstract*— In practical synthetic aperture radar (SAR) target detection applications, it is often encountered that the training and testing data come from different SAR sensors, leading to a decline in the SAR target detection performance. Although domain adaptation methods can achieve model generalization through feature transfer, the change of scattering characteristics for the same target and the difference in feature distribution, caused by cross sensor, cannot be ignored in SAR images. It is inevitable to lead to the escalation of the offset in the bounding box regression and deviation of the feature alignment. To address these issues, a cross-sensor SAR image target detection method based on the dynamic feature discrimination and center-aware calibration is proposed. Based on the domain adaptation framework, initially, a dynamic feature discrimination module (DFDM) is introduced to address the exacerbated offset in the regression. A bidirectional spatial feature aggregation (BSFA) mechanism is employed to aggregate features in both horizontal and vertical directions, and a multiscale structure is adopted to enhance the scattering and semantic features, which can dynamically constrain the target position while improving target discrimination capability. Then, the center-aware calibration module (CACM) is designed to address the alignment deviation in feature transfer. The target salience relationship is modeled based on the distance between different positions and the target center to suppress the background clutter interference. The perception center of the target is focused by combining the centerness map and classification map, which can calibrate the domain-invariant features and alleviate misalignment. Finally, the proposed method is tested on two datasets, MiniSAR and FARAD, and compared with the latest domain adaption methods. Both mAP and $F1$ values have improved by more than 6%–20%, verifying the effectiveness of the proposed method.**

***Index Terms*— Center-aware calibration, cross-sensor, domain adaptation, dynamic feature discrimination, synthetic aperture radar (SAR) target detection.**

I. Introduction

SYNTHETIC aperture radar (SAR), known for its all-weather and all-day capabilities, has been extensively utilized in various domains. It actively acquires targets by emitting electromagnetic waves, playing a pivotal role in military and civilian applications such as ocean monitoring, disaster assessment, and resource management [1], [2], [3].

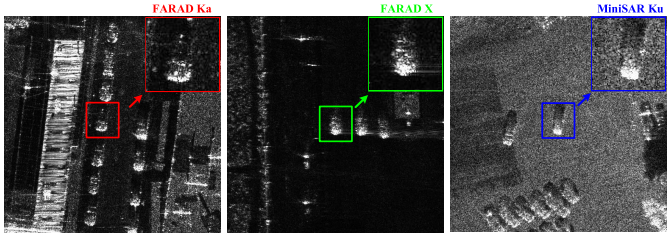


Fig. 1. SAR vehicle targets under different sensors.

The traditional SAR target detection method mainly includes the constant false alarm rate (CFAR) algorithm and its variants algorithms [4], [5], [6], [7], [8], which leverages the distribution of targets and background clutter to distinguish targets from the background. However, these methods suffer from limited detection accuracy and a propensity for high false alarms. In recent years, with the significant advancements of deep learning in the field of remote sensing image interpretation, deep neural networks have been increasingly applied to SAR target detection and recognition. Researchers have studied and designed numerous algorithms on targets, such as aircraft [9], [10], [11], [12], ships [13], [14], [15], [16], and vehicles [17], [18], [19]. Many of them have achieved outstanding performance and made significant progress. However, most existing research on the SAR target detection focuses on scenarios, where the training and testing data are from the same sensor.

In recent years, as the variety of SAR sensors has gradually increased, the actual operation conditions often involve training and testing data from different sensors, leading to significant differences in data distribution. This poses a considerable challenge for SAR target detection. Different SAR sensor platforms have different system parameters, such as radar frequency, beamwidth, orbital height, and observation angle, resulting in significant differences in target characteristics and feature distributions. In terms of target characteristics, as shown in Fig. 1, which displays the vehicle data from three different sensors, the data from three distinct sensors show obvious differences in gray intensity, image style, and scattering structure. Regarding feature distribution, as shown in Fig. 2, which presents the t-distributed stochastic neighbor

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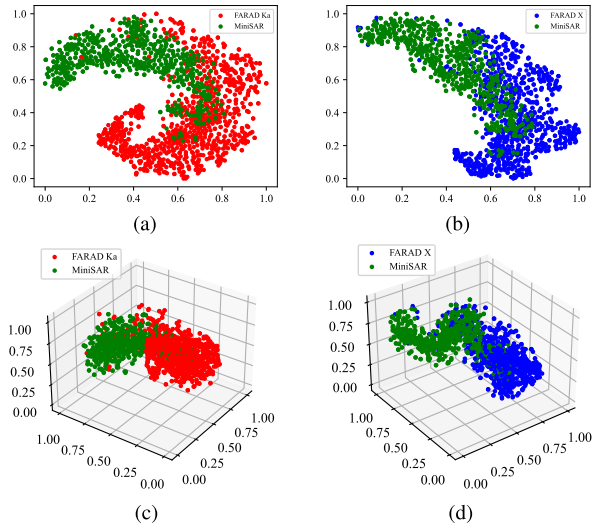


Fig. 2. Feature distribution of datasets with different sensors. (a) and (b) 2-D t-SNE visualization maps. (c) and (d) 3-D t-SNE visualization maps.

embedding (t-SNE) visualization of data from different sensors, there are significant differences in the feature distribution among data from different sensors. Therefore, when the sensor of the test data is inconsistent with that of the training data, the performance of SAR target detection will significantly decline, and its generalization ability will be greatly reduced. A common solution is to annotate new sensor data and retrain the model. However, since the SAR image interpretation requires certain expert knowledge, it is relatively difficult to interpret. Moreover, annotation is time-consuming and labor-intensive, making it challenging to conduct accurate and efficient annotation. Therefore, how to enhance the SAR target detection performance under conditions where the sensors of testing and training data are inconsistent is an urgent problem to be solved.

Recently, researchers have proposed a domain adaptation method [20], [21], [22] in the field of machine learning. This method can map the data from the source domain and the target domain into the same feature space and align the distributions of the two domains by minimizing the distance between them using appropriate metric criteria, thus achieving the transfer from the source domain to the target domain. Domain adaptation was initially applied to image classification tasks [23], [24], [25]. As this method evolved, it gradually began to be used for object detection tasks and is currently widely applied in the fields of computer vision and optical remote sensing. Chen et al. [26] enhanced the Faster-RCNN framework and proposed a cross-domain target detection method based on image-level and instance-level feature alignment, achieving superior results on various computer vision-related datasets. Hsu et al. [27] improved cross-domain detection performance by classifying and aligning each pixel of the extracted feature map at the global level. Chen et al. [28] introduced a cross-domain detection method in conjunction with CycleGAN, employing an image-to-image conversion module to create interpolated images, thus enhancing model adaptability. Li et al. [29] integrated graph structure with domain adaptation, improving the generalization of object detection between different domains through graph matching and local clustering. Wang et al. [30] proposed a domain adaptation method based on the sequence feature alignment, enhancing the cross-regional capability of detection transformers by utilizing sequence feature alignment based on the domain query and word-by-word sequence feature alignment.

In the field of cross-domain SAR image target detection, some domain adaptation algorithms have emerged. These mainly include methods of multifeature alignment [31], [32], [33], [34], pseudoimage generation [35], domain-invariant feature decomposition [36], and transformer-based method. However, domain adaptation [37] in the SAR field is still in its infancy, especially in cross-sensor SAR target detection with significant challenges to be solved. On the one hand, changes in sensors make it no longer ignore the alterations in the characteristics of targets on images, which can exacerbate the offset of bounding box regression. The differences in SAR sensors are accompanied by various radar parameter discrepancies, including radar frequency, wavelength, and observation angle. These may cause changes in the local backscattering intensity of the target, shifts in scattering centers, and variance in shape contours. Such changes in target characteristics can lead to a more severe offset in the regression of the target bounding box. On the other hand, changes in sensors can lead to differences in the feature distribution, resulting in alignment deviation during feature transfer. SAR images are generated from radar echoes. Variations in sensors can produce different scattering echoes, leading to significant distinctions in feature distribution. The positive samples of targets and the negative samples of the background can be confused inside the bounding box, causing alignment deviation during feature transfer. Moreover, the distribution of background clutter has large differences, which can generate a large amount of redundant information and lead to alignment deviation in alignment.

To address the above challenges, a novel cross-sensor SAR target detection method is introduced, combining dynamic feature discrimination module (DFDM) and center-aware calibration module (CACM). In DFDM, a bidirectional spatial feature aggregation (BSFA) mechanism is employed to aggregate features horizontally and vertically to realize the dynamic adjustment of direction perception. Also, a multiscale structure is adopted to enhance the scattering and semantic features of the target. In CACM, the distance between different positions and the target center is modeled to characterize the salience relationship. In addition, the centerness map and classification map are combined to focus on the perception center of the target. The main contributions are as follows.

1) A cross-sensor domain adaptation SAR image target detection method is proposed, which can significantly improve the generalization ability of the SAR target detection model under different sensor conditions.
2) The DFDM is designed to tackle the escalation of the offset in the bounding box regression. By adopting the BSFA mechanism and the multistructure, the target position can be precisely constrained, and the discrimination capability can be significantly improved.

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3) The CACM is introduced to address the feature alignment deviation in the feature transfer. In this module, the salience relationship is characterized and the perception center is focused, which can help calibrating the domain-invariant features and alleviate misalignment caused by negative samples.

II. Related Works

A. Deep Learning-Based SAR Target Detection

At present, there are many excellent object detection algorithms based on deep learning, including single-stage and two-stage object detection methods. The classical methods of single-stage target detection mainly include RetinaNet [38], YOLO [39], and SSD [40]. The two-stage target detection method mainly includes R-CNN [41] and Faster-RCNN [42]. The object detection method based on deep learning has gradually become a hot topic in the field of image interpretation and has also obtained great performance in the field of SAR object detection. Pan et al. [43] introduced scattering features into the deep learning object detection framework to obtain high detection performance on ship images in large scenes. Li and Lang [44] proposed a new method for rotating detection of SAR ship targets. The mixed convolutional channel attention module is embedded in the backbone network to enhance the detection performance by reweighting feature maps and all channels that represent ship features. Zhou et al. [45] added a transformer module to the backbone and improved the ability to discriminate critical features, which reduced the number of parameters and showed excellent detection performance. Wang et al. [46] designed the multiscale pyramid attention module in the network to enhance the feature representation, which can achieve the excellent performance of efficient lightweight detection in three SAR ship datasets. Zhou et al. [47] proposed a detection network combined with the Swin transformer. They designed a novel feature aggregation module and mix-attention mechanism to solve the complex backgrounds and small target problems in SAR target detection. However, these deep learning-based detection methods are only trained and tested on the data of the same sensor, without considering the problem of cross-sensor and different distributions.

B. Domain Adaptation SAR Target Detection

To improve the adaptability of the detection model, the method of unsupervised domain adaptation has been widely used by Vettoruzzo et al. [21]. The research on the domain adaptive SAR target detection mainly includes optical domain to SAR domain adaptation and SAR domain to SAR domain adaptation. At present, there have been some breakthroughs in the field of optic-to-SAR domain adaptation detection. Guo et al. [31] proposed an improved domain adaptive method of Faster-RCNN, which performed adversarial training on ROI layer features to realize the optic-to-SAR adaptation. Shi et al. [32] integrated different fusion strategies to make the generator pay more attention to the texture and structure features of SAR images, helping to generate the transition domain images with more realistic visual effects. Zhang et al. [35] introduced a pseudoimage generation method based on the Fourier transform, which combined phase spectrum and amplitude spectrum to achieve different feature alignments. Huang et al. [48] added the speckle noise to the optical images to alleviate the visual difference to the SAR domain. The spatial information and the instance-invariant features are combined to enhance the domain adaptation performance of rotating targets. Dong et al. [49] designed and integrated several attention mechanisms to enhance the interpretability of contextual understanding, which can leverage the complementary advantages of different domains. Recently, some studies have emerged in the field of SAR-to-SAR domain adaptive target detection. Zhao et al. [36] proposed a domain adaptation method to decompose the extracted features into domain-invariant features and domain-specific features, which can improve the cross-domain detection capability of SAR ship targets. Shi et al. [33] integrated multilayer pixel domain adaption and iterative pseudotagging methods to improve the cross-sensor target detection capability on vehicle datasets, which can make excellent detection performance. Liu et al. [34] introduced multilevel semantic calibration and uncertainty-aware guided alignment into the domain adaptation network to capture the intrinsic features of target, showing high accuracy on three ship datasets. Zhao et al. [37] combined transformer structure and domain knowledge to realize the sequence feature alignment, which can improve the generalization of SAR ship target detection. Qin et al. [50] employed the geometry-aware clustering assignment and unbalanced optimal transport to map the prototypes and reduce the discrepancies of feature distribution in different domains. These proposed methods can reduce the feature distribution of different domains and can achieve great generalization performances. However, most of the methods fail to fully account for the scattering features of SAR targets and the feature distribution in cross-sensor tasks, which are crucial for dynamically characterizing target features and domain-invariant features. This limitation can lead to the exacerbation of the offset on the bounding box and the misalignment during feature alignment. Also, it is susceptible to interference from the background clutter, resulting in an increased number of generation of redundant features and false alarms. The approaches in this field are still in the infancy.

III. Methodology

In this section, the proposed cross-sensor SAR target detection algorithm is described in detail. Problem description is introduced in Section III-A. The DFDM is described in Section III-B. The CACM is shown in Section III-C. The cooperative loss optimization strategy is introduced in Section III-D. The overall framework of the proposed method is shown in Fig. 3.

A. Problem Description

For cross-sensor SAR domain adaptive target detection, the detection model trained on the source domain needs to be adapted to the target domain. Through the feature transfer, the target feature space of these two domains becomes closed and the generalization ability of detection is improved.

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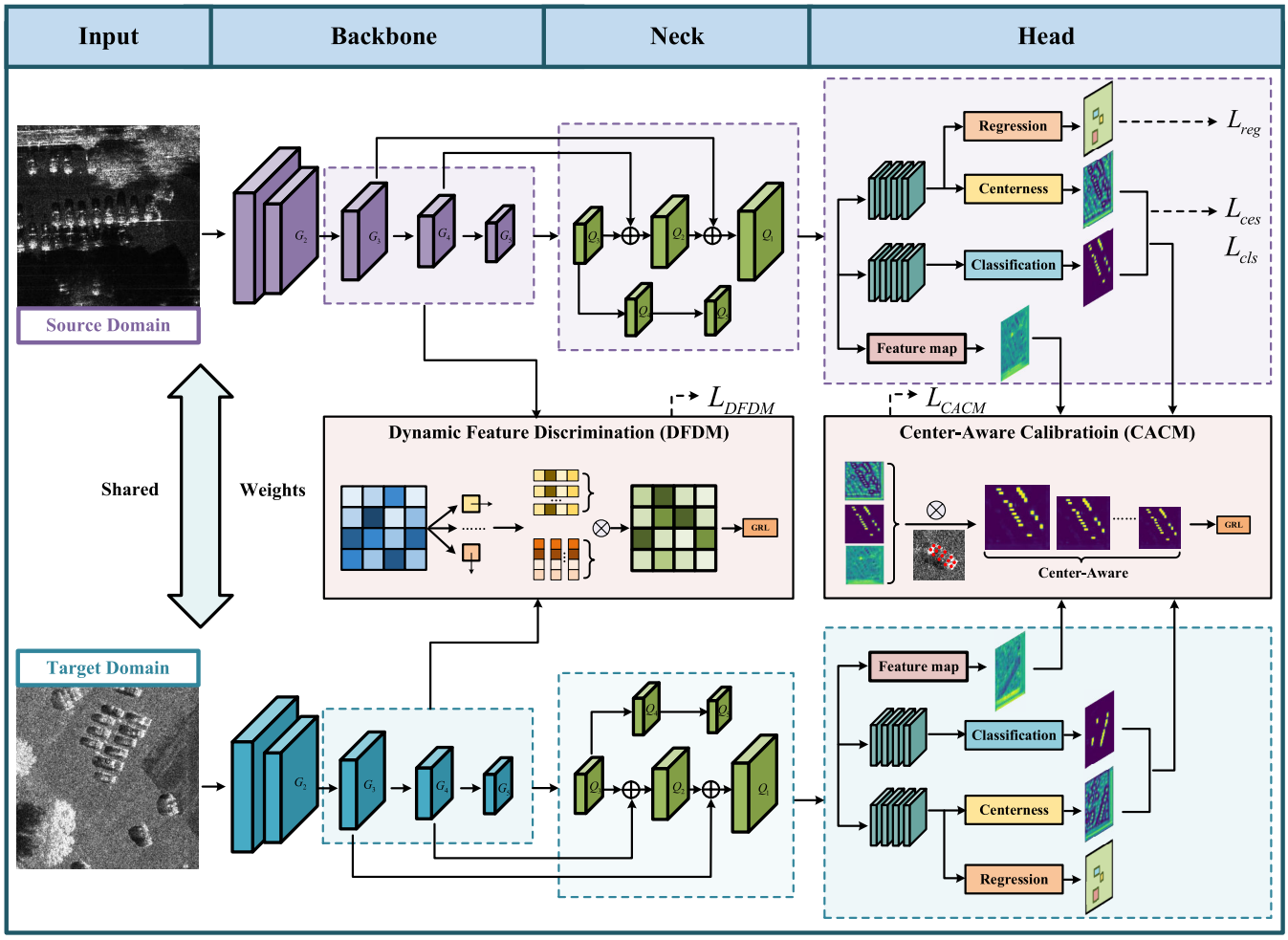


Fig. 3. Overall structure of the proposed cross-sensor SAR target detection method based on the dynamic feature discrimination and center-aware calibration. The framework includes the input of two data domains. G_1–G_5 and Q_1–Q_5 are the feature extraction layers of the backbone network. GRL is the gradient reversal layer. The DFDM and CACM are the two main modules in the proposed method.

Assuming that the source domain data with complete annotation is defined as $D_s = \{(x_k^s, y_k^s, l_k^s)\}_{k=1}^{N_s}$, where N_s denotes the number of samples source domain contains, x_k^s represents the source domain sample, and y_k^s represents the category label. In addition, the unlabeled target domain data is defined as $D_t = \{(x_k^t)\}_{k=0}^{N_s}$, where x_k^t represents the target domain sample. k denotes the domain label, with 1 and 0 being the source domain and target domain, respectively. The purpose of the proposed method is to obtain the label y_k^t and bounding box l_k^t of the target domain data through the feature transfer between the source domain and the target domain.

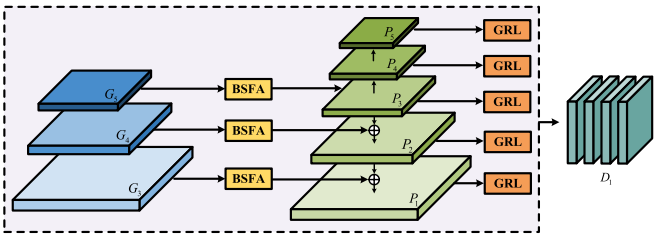


Fig. 4. Structure of DFDM. BSFA denotes the bidirectional spatial feature aggregation mechanism.

B. DFDM

The change of scattering characteristics for the same target can lead to the escalation of the offset in the bounding box regression. To address the escalation of the offset in cross-sensor SAR target detection tasks, the algorithm framework introduces a DFDM. By employing a BSFA mechanism to aggregate features in both horizontal and vertical directions, the module can dynamically adjust the weights of feature maps based on the direction perception information, capturing pixel-level correlations and precisely constraining target detection boxes to better preserve position information of targets. The discrimination network incorporates a multiscale structure that integrates low-level and high-level features, enhancing the discrimination of SAR target texture details and semantic features in both low and high dimensions, thereby mitigating the loss of scattering information of targets due to the increased dimension in the discrimination network. The structure of the module is shown in Fig. 4.

First, the source domain data x_i^s with full annotation and the target domain data x_i^t with no annotation are, meanwhile, input into a shared feature extractor G. The features are then

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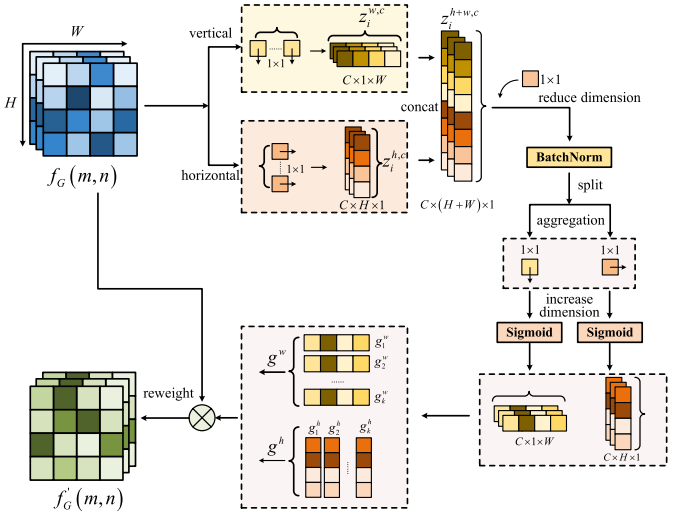


Fig. 5. Specific structure of BSFA mechanism.

extracted as f_G

$$f_G = G(x). \tag{1}$$

The feature extraction backbone includes five convolution layers, with each layer G_i producing a feature map denoted by f_{G_i}. To bolster the accuracy of object detection and refine localization, a BSFA mechanism is introduced, which can operate between the G_i and P_i layers. The structure of the BSFA mechanism is shown in Fig. 5. This mechanism aggregates features from both horizontal and vertical directions, allowing the network to concentrate on the pivotal features within the feature map. By dynamically adjusting the weights of various regions based on the significance of the target areas, the BSFA mechanism effectively captures the interpixel relationships, thereby providing a spatially accurate localization of different targets. After integrating the G_i layer with BSFA, the P_i layer can obtain rich position information. This advancement is particularly valuable for the field of cross-sensor SAR object detection, where the precision (PR) of localization is of utmost importance. The BSFA mechanism can not only select more critical features but also mitigate the position ambiguity that results from the conventional unidirectional convolution, thereby enhancing the overall performance of the detection algorithm.

The BSFA mechanism initially decomposes global pooling into two subpooling operations along two spatial dimensions. It performs channel encoding and information integration along the vertical and horizontal dimensions of the feature map, respectively, which can generate two directional sensing feature maps containing different position information. Consequently, the feature map outputs at locations h and w for channel c can be denoted by $z_i^{h,c}$ and $z_i^{w,c}$

$$z_i^{h,c}(h) = \frac{1}{W}\sum_{0\le w<W} f_{G_i}(h,w) \tag{2}$$

$$z_i^{w,c}(w) = \frac{1}{H}\sum_{0\le h<H} f_{G_i}(h,w) \tag{3}$$

where h and w represent the height and width, respectively. The transformations can aggregate features along two orthogonal spatial axes, generating a pair of direction-sensitive feature maps $z_i^{h,c} \in \mathbb{R}^{C\times H\times 1}$ and $z_i^{w,c} \in \mathbb{R}^{C\times 1\times W}$, where C represents the number of categories.

These transformations are conducive to the ability to capture long-range dependencies across diverse spatial orientations, highlighting regions of significance within the targets, and preserving precise position information. The enhancement of position can help to localize the targets accurately and assign weights dynamically to various regions of the feature map in the network. The features aggregated from two orthogonal directions provide a more comprehensive reflection of the position information of the target, thereby improving the accuracy of target detection from multiple directions. Following this, the feature maps pooled in two distinct directions are concatenated to produce an enriched feature map z^{h+w}

$$z^{h+w} = \text{concat}\left(z^h, z^w\right) \tag{4}$$

where the feature maps of two directions are combined and embedded with position information of vertical and horizontal directions.

Then, the 1×1 convolutional kernel is employed to perform the dimension reduction, producing a feature map denoted by f. Subsequently, the feature map f undergoes a split operation, which divides it along the horizontal and vertical directions into two separate components, f^h and f^w, respectively. These components are then subjected to additional 1×1 convolutional operations of F_h and F_w, which serve to increase the dimension of the feature maps from vertical and horizontal dimensions. The final weights can be denoted by g^h and g^w

$$g^h = \sigma\left(F_h(f^h)\right) \tag{5}$$

$$g^w = \sigma(F_w(f^w)) \tag{6}$$

where the sigmoid activation function, denoted by σ, is applied to these weights to introduce nonlinearity and to ensure that the output values are confined within the range from 0 to 1.

The obtained weights g^h and g^w, which are derived for the horizontal and vertical directions, respectively, are utilized to dynamically adjust the feature map output from the previous layer. Different pixel positions on the images are denoted by (m,n). The obtained weights on the layer G_i can be represented as $g_{G_i}^h$ and $g_{G_i}^w$. After the BSFA, this process involves elementwise multiplication of these weights with the feature map $f_{G_i}(m,n)$, resulting in a refined feature map $f'_{G_i}(m,n)$

$$f'_{G_i}(m,n) = f_{G_i}(m,n)\cdot g_{G_i}^h(m)\cdot g_{G_i}^w(n) \tag{7}$$

where i takes the value of 3–5. This method allows for a more focused representation of features, which can emphasize the significance of certain regions based on the relevance of direction perception and achieve adaptive channel information weighting.

After the BSFA mechanism, the feature map f_{G_3}, f_{G_4}, and f_{G_5} can be assigned adaptive dynamic weights with position information and can be denoted by f'_{G_3}, f'_{G_4}, and f'_{G_5}.

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TABLE I
SPECIFIC STRUCTURE OF THE DISCRIMINATOR

Layer	Function	Specific Setting
1	Convolution	256×3×3, stride 1
2	GroupNorm, ReLU	—
3	Convolution	256×3×3, stride 1
4	GroupNorm, ReLU	—
5	Convolution	256×3×3, stride 1
6	GroupNorm, ReLU	—
7	Convolution	256×3×3, stride 1
8	GroupNorm, ReLU	—

A feature pyramid is used to enhance the scattering feature information of the target in the network, and the feature map of different scales f_{P_i} is obtained.

Subsequently, the feature maps of different scales, denoted by f_{P_i}, are fed into the gradient reversal layer (GRL). Following this, the feature maps are put into the feature discriminator D_1, which serves the dual purpose for domain classification and feature discrimination. The output from the discriminator is a domain classification map that mirrors the dimensions of the feature map f_{P_i}. Each element within this map corresponds to the domain label at the respective position on the feature map. For clarity, the domain labels y for the source and target domains are assigned as 1 and 0, respectively,

$$y = \begin{cases} 1, & \text{Source Domain} \\ 0, & \text{Target Domain.} \end{cases} \tag{8}$$

The optimization of this discrimination process is achieved by minimizing the binary cross-entropy loss, which can effectively enhance the ability to differentiate between domains.

The feature adversarial discriminator D_1 mainly consists of four convolutional layers. The specific structure of the discriminator is shown in Table I. Due to the feature pyramid subsampling layer of the feature extractor, a large amount of semantic feature information of SAR targets may be lost, resulting in inaccurate discrimination. Therefore, to compensate for the loss of scattering information, a multiscale feature discrimination method is designed. The feature maps of different scales are input into the discriminator to reduce the loss of scattering information. The multiscale structure can help obtain multilevel discrimination loss, and can more comprehensively represent the feature difference between the source domain and target domain. The feature map of $f_{P_i}^s$ and $f_{P_i}^t$ represent the source domain and the target domain, respectively. The dynamic feature discrimination loss obtained by different scale layers of P_i is expressed as $\mathcal{L}_{P_i}$

$$\begin{aligned} \mathcal{L}_{P_i} = &- \sum_{m,n} y \log\left(D_1\left(f_{P_i}^s(m,n)\right)\right) \\ &+ (1-y)\log\left(1 - D_1\left(f_{P_i}^t(m,n)\right)\right) \end{aligned} \tag{9}$$

where (m, n) denote different positions on the feature map and y represents the domain label.

Then, the losses of different scales are combined to characterize the discrimination capability more accurately. The total loss of DFDM is obtained through the combination of losses in different scales, which is denoted by $\mathcal{L}_{\text{DFDM}}$

$$\mathcal{L}_{\text{DFDM}} = \sum_{i=1}^{K_d} \mathcal{L}_{p_i} \tag{10}$$

where K_d represents the total number of the layers involved in feature discrimination.

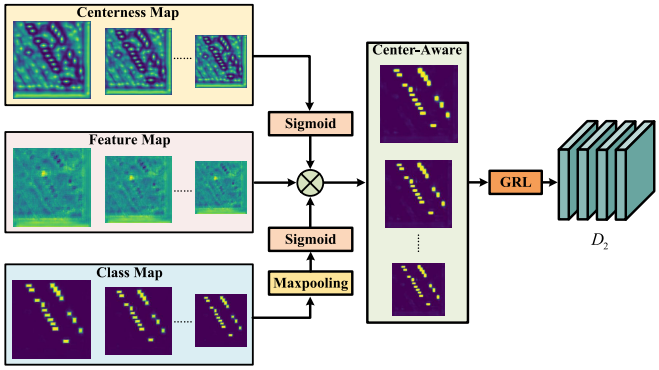


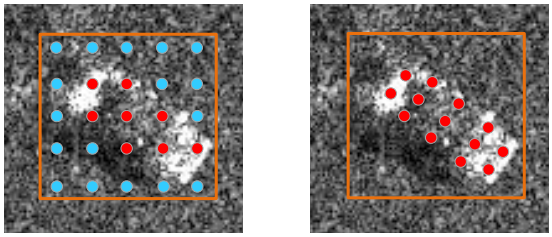
Fig. 6. Specific structure of CACM.

C. CACM

The difference in the feature distribution can cause the deviation of the feature alignment. To address the alignment deviation in feature transfer caused by background clutter inside the bounding boxes, a CACM has been designed. The structure of the CACM is shown in Fig. 6. This module enhances the representation of critical scattering components of targets based on the distance between the target center and different positions. It can help suppress the background areas and mitigate the interference from surrounding clutter, thereby reducing false alarms. The two different alignment methods are shown in Fig. 7. The traditional instance-level alignment is performed on positive and negative samples, as shown in Fig. 7(a), which is prone to misalignment and redundant information, leading to inaccurate feature discrimination. By integrating the centerness map, classification map, and feature map, the module characterizes the perception center of targets and focuses on the target area, as shown in Fig. 7(b). This approach can enhance the salience of targets in the image and calibrate the domain-invariant intrinsic features of targets. This can help the network concentrate on the strong response center areas and alleviate the misalignment introduced by negative samples inside the bounding boxes.

The features extracted by the feature extractor are initially directed into a feature pyramid to enhance the capacity for capturing scattering information. Within the feature pyramid, the backbone network can extract feature maps of diverse scales, which are denoted by f_{Q_i}. These feature maps are then routed through two distinct fully convolution networks with four layers, which can produce the classification output map $M_{\text{cls}} \in \mathbb{R}^{H\times W\times C}$ and the centerness output map $M_{\text{ces}} \in \mathbb{R}^{H\times W}$, respectively. The values within the classification output map range from 0 to 1. Each position on the classification output and the centerness output map indicates the corresponding classification result and centerness score,

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(a) (b)

Fig. 7. Two different methods of target feature alignment. (a) Alignment method that focuses on both positive and negative regions inside the bounding box. (b) Proposed center-aware calibration method that focuses on the target region.

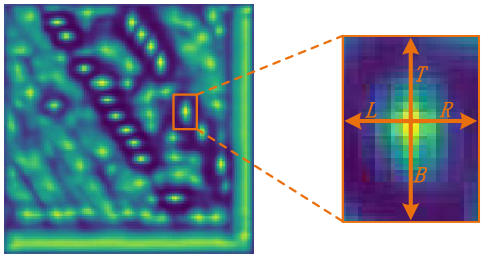


Fig. 8. Calculation method of the centerness map.

respectively. Furthermore, the model derives three key losses, that is, the classification loss $\mathcal{L}_{\text{cls}}$, the centerness loss $\mathcal{L}_{\text{ces}}$, and the position regression loss of the target $\mathcal{L}_{\text{reg}}$. These losses are pivotal for refining the performance in accurately classifying categories, calculating centerness, and regressing target positions.

The centerness map M_{ces} signifies the centerness distribution of the target [51]. It quantifies the distance between the attention center of the network and the image pixel coordinates (m, n). The distances to the left, right, top, and bottom boundaries of the target anchor are denoted by d_l, d_r, d_t, and d_b. The centerness score can be calculated as centerness(d_l, d_r, d_t, d_b)

$$\text{centerness}(d_l, d_r, d_t, d_b) = \sqrt{\frac{\min(d_l, d_r)}{\max(d_l, d_r)} \cdot \frac{\min(d_t, d_b)}{\max(d_t, d_b)}}. \tag{11}$$

The centerness score diminishes as the anchor point moves further from the target center, and conversely, it escalates as the anchor point approaches the target center. This relationship is graphically depicted in Fig. 8. The centerness score effectively assigns weights to the distances from the center of the target. It aids in differentiating between the foreground and background regions, where the foreground areas receive higher scores and the background areas receive lower scores. This approach mitigates the influence of background clutter and reduces the misalignment caused by negative samples to some extent. By reducing the impact of background clutter, distinguishing the background from the target, and enhancing the PR of target alignment, this operation can improve the overall effectiveness of cross-sensor SAR target detection.

The classification map M_{cls} can display the classification results of different positions on the image and classify different pixel areas on the image. It can ignore the area with false classification and remove false alarms, such as trees, buildings, and roads, to obtain the image area more accurately and efficiently, and then align the feature of the target area accurately. The value of the classification result ranges from 0 to 1. To reduce the dimension, Maxpooling is operated on the classification map M_{cls} to obtain M'_{cls}

$$M'_{\text{cls}} = \text{MaxPooling}(M_{\text{cls}}). \tag{12}$$

The use of centerness map M_{ces} can better highlight the center position of the target and the strong response area of the target in the SAR image. It can model the intensity of the target structure according to the response center. Different weights of classification map M_{cls} can determine the potential target location, which can more effectively characterize the region possessing domain-invariant characteristics and accentuate the critical feature areas. To concentrate more on the strong scattering regions of the target and focus on the center of the target, classification map M_{cls} and centereness map M_{ces} are multiplied to calibrate the domain-invariant features. Before the multiplication, M'_{cls} and M_{ces} each pass through a sigmoid layer. Thus, we can obtain the center classification map M_{cc}

$$M_{\text{cc}} = \sigma\left(\sigma\left(M'_{\text{cls}}\right) \odot \sigma(M_{\text{ces}})\right). \tag{13}$$

Since the weights on the feature map of the network can highlight the critical areas that the network focuses on, the center classification map of the source domain and target domain is multiplied by the feature map to get the center-aware map f^s_{CA} and f^t_{CA}

$$f^s_{\text{CA}} = M^s_{\text{cc}} \odot f_s \tag{14}$$

$$f^t_{\text{CA}} = M^t_{\text{cc}} \odot f_t \tag{15}$$

where M^s_{cc} and M^t_{cc} represent the center classification map of the source domain and the target domain, respectively. f_s and f_t represent the feature map of different domains.

The result of the center-aware map represents the classification weight score of different regions from the target center. A higher weight score signifies the importance of a feature, indicating greater scattering intensity and richer scattering information. This enhanced representation is crucial for characterizing the scattering structure and texture features of the target more effectively. It allows for a more detailed understanding of local changes and distribution within the target, thereby highlighting the center response region.

Furthermore, the domain-invariant characteristics of the source and target domains have been further established to mitigate the impact of negative sample background areas and enhance the salience of targets with weak scattering intensity. This approach can ensure that the focus of the model is directed toward the center features of the target and can reduce the interference from background noise. By emphasizing the invariant features across different domains, the model can more effectively characterize the scattering structure and texture features of the target, attend to local changes and distribution within the target, and highlight the center response region of the target.

After the center-aware maps of the source domain f^s_{CA} and the target domain f^t_{CA} are put through the GRL, the

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center-aware calibration is realized by the feature discriminator D_2. The specific structure of the discriminator is the same as D_1, as shown in Table I. The center-aware calibration loss of f_{Q_i} can be represented as $\mathcal{L}_{\mathrm{CA}_i}$

$$\mathcal{L}_{\mathrm{CA}_i} = -\sum_{m,n} y \log\left(D_2\left(f^s_{\mathrm{CA}_i}(m,n)\right)\right) + (1-y)\log\left(1 - D_2\left(f^t_{\mathrm{CA}_i}(m,n)\right)\right). \tag{16}$$

To enhance the accuracy of the center-aware calibration, the center-aware calibration losses of different scales are combined. The sum of the obtained center-aware calibration losses can be expressed as $\mathcal{L}_{\mathrm{CACM}}$

$$\mathcal{L}_{\mathrm{CACM}} = \sum_{i=1}^{K_c} \mathcal{L}_{\mathrm{CA}_i} \tag{17}$$

where K_c denotes the number of layers involved in the CACM.

Algorithm 1 Overall Procedure of the Detector

Input:
Labeled source images: D_s;
Unlabeled target images: D_t;
Output:
An adaptive detector.

```
Initialize the model parameters randomly;
while not converged do
    Sample a mini-batch from D_s and D_t;
    Extract the features of images in different domains;
    Calculate the detection loss L_cls, L_reg and L_ces;
    Send the feature maps f^s_{P_i} and f^t_{P_i} of multilayer into
    the dynamic feature discrimination module;
    Calculate dynamic discrimination loss L_DFDM;
    Send the feature maps f^s_{CA_i} and f^t_{CA_i} of multilayer
    into the center-aware calibration module;
    Calculate center-aware calibration loss L_CACM;
    Merge different Loss to characterize the difference
    L_total = L_cls + L_reg + L_ces + λL_DFDM + μL_CACM
    Optimize the L_total to realize knowledge transfer;
end
```

D. Cooperative Loss Optimization Strategy

To characterize the model loss more comprehensively and to improve the capability of feature discrimination, the cooperative loss optimization strategy is adopted. Since the discrimination loss under a single scale cannot fully realize the feature discrimination, the dynamic feature discrimination loss $\mathcal{L}_{\mathrm{DFDM}}$ and center-aware loss $\mathcal{L}_{\mathrm{CACM}}$ obtained at different scales are combined to capture the targets of different sizes and reflect the difference between target domain and source domain more precisely.

In addition, to characterize the feature map and the location of the target precisely, the classification loss $\mathcal{L}_{\mathrm{cls}}$, location regression loss $\mathcal{L}_{\mathrm{reg}}$, and centerness loss $\mathcal{L}_{\mathrm{ces}}$, obtained by classification and regression of source domain data, are calculated in the total loss. The overall procedure of the detector can be concluded as Algorithm 1. The $\mathcal{L}_{\mathrm{cls}}$ is calculated based on the focal loss [52]. It can be formulated as follows:

$$\mathcal{L}_{\mathrm{cls}} = \sum_{(m,n)\in S_p} -\alpha(1-R_{\mathrm{cls}})^{\gamma}\log(R_{\mathrm{cls}}) + \sum_{(m,n)\in S_n} -(1-\alpha)(R_{\mathrm{cls}})^{\gamma}\log(1-R_{\mathrm{cls}}) \tag{18}$$

where S_p and S_n denote the regions of positive samples and negative samples, respectively. R_{cls} represents the prediction of the classification in different positions. α and γ refer to the modulating factor.

$\mathcal{L}_{\mathrm{reg}}$ denotes the loss of the location regression, which can be measured based on IOU loss [53]. It can be defined as follows:

$$\mathcal{L}_{\mathrm{reg}} = \sum_{(m,n)\in S_u} (1 - \mathrm{IOU}(R_{\mathrm{bbox}}, T_{\mathrm{bbox}})) \tag{19}$$

where S_u is the union of the positive sample regions and the invalid sample regions. R_{bbox} and T_{bbox} refer to the prediction and the ground truth of the bounding box in different positions, respectively.

$\mathcal{L}_{\mathrm{ces}}$ represents the loss of the centerness, which is defined according to the binary cross entropy [51]. It can be formulated as follows:

$$\mathcal{L}_{\mathrm{ces}} = \sum_{(m,n)\in S_u} \mathrm{BCE}(R_{\mathrm{ces}}, T_{\mathrm{ces}}) \tag{20}$$

where R_{ces} and T_{ces} refer to the prediction and the ground truth of the bounding box in different positions, respectively.

By combining these different losses, $\mathcal{L}_{\mathrm{total}}$ can be calculated as follows:

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{cls}} + \mathcal{L}_{\mathrm{reg}} + \mathcal{L}_{\mathrm{ces}} + \lambda\mathcal{L}_{\mathrm{DFDM}} + \mu\mathcal{L}_{\mathrm{CACM}} \tag{21}$$

where λ and μ denote the weights of the DFDM and the CACM, respectively. The target detection algorithm is optimized by the iterative optimization of $\mathcal{L}_{\mathrm{total}}$.

IV. Experiments and Results

In this section, we introduce the datasets of the experiment, describe the details of the experimental setup, compare the results of the method with the state-of-the-art methods, and analyze the ablation experiments.

A. Datasets Description

In this article, three datasets from different imaging sensors are used to verify the performance of the algorithm. The SAR data of MiniSAR and FARAD obtained by Sandia National Laboratories in 2005 and 2015, respectively, are used to verify the method in this article [19]. The imaging sensor band of MiniSAR data is the Ku band. The shooting location is in the suburbs of Kirtland Air Force Base. However, the band of FARAD data is the Ka band and X band. The shooting location of FARAD is in the densely built urban area of the University of New Mexico. These two kinds of datasets cover three kinds of imaging bands, among which FARAD and MiniSAR adopt completely different sensors. The samples of different datasets are shown in Fig. 9. The specific information of the data is shown in Table II.

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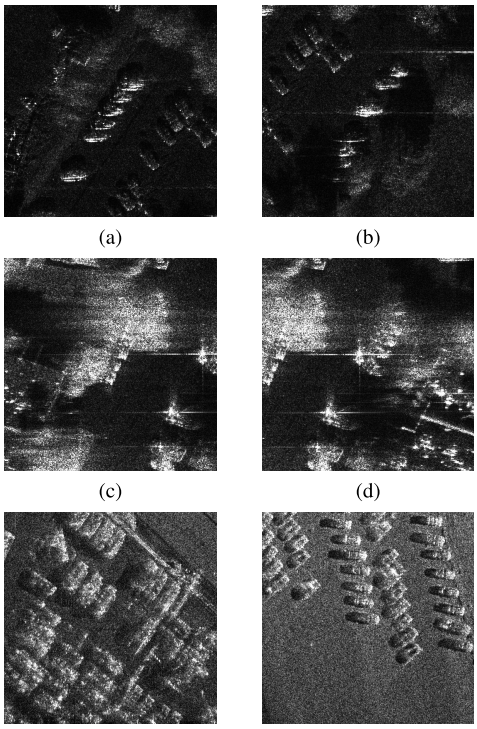


Fig. 9. Different samples of three vehicle datasets. (a) and (b) Samples from FARAD Ka. (c) and (d) Samples from FARAD X. (e) and (f) Samples from MiniSAR.

TABLE II
COMPARISONS OF IMAGING PARAMETERS BETWEEN THREE DATASETS

Datasets	FARAD Ka	FARAD X	MiniSAR
Wave	Ka	X	Ku
Time	2015.08	2015.10	2005.05
Bandwidth	5 GHz	3 GHz	3 GHz
Pitch Angle	26°-34°	26°-34°	26°-29°
Resolution	0.1 m	0.1 m	0.1 m
Center Frequency	35.6 GHz	9.6 GHz	16.8 GHz
Max Observation range	6 km	12 km	8 km

B. Experimental Setup

The whole experiment is trained on Intel Core(R) i9-14900KF Intel Core(R), NVIDIA GeForce RTX 4070Ti Super and Ubuntu 20.04. The initial learning rate is 0.005, weight decay is 0.0001, and IOU threshold is 0.5. The maximum iteration is 20 000. The weights of losses λ and μ are both set to 0.5. TP is defined as the number of correctly classified positive samples. FP is defined as the number of incorrectly classified positive samples. FN is defined as the number of incorrectly classified negative samples. The four indicators of recall (RE), PR, mAP, and $F1$ are used in the evaluation of the detection performance.

RE represents the ratio of correctly predicted targets by the network to all ground-truth targets. RE is defined by

$$\mathrm{RE} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{22}$$

PR represents the ratio of correctly predicted targets by the network to all predicted targets. PR is defined by

$$\mathrm{PR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}. \tag{23}$$

The mAP represents the mean average PR of detection results. The mAP is defined by

$$\mathrm{mAP} = \frac{1}{N} \sum_{N} \left[\int \mathrm{PR}(\mathrm{RE}) \mathrm{d}(\mathrm{RE}) \right]. \tag{24}$$

$F1$ measures the overall detection performance, which is jointly determined by PR and RE. $F1$ is defined by

$$F1 = 2 \cdot \frac{\mathrm{RE} \cdot \mathrm{PR}}{\mathrm{RE} + \mathrm{PR}}. \tag{25}$$

C. Comparisons With State-of-the-Art Methods

To validate the performance of our proposed method, we conducted a comparative analysis of some recent deep learning methods with great performance and state-of-the-art domain adaptation target detection methods. The deep learning detection methods include Faster-RCNN [42], DINO [54], YOLOF [55], and RTMDet [56]. The domain adaptation detection methods include DAF [26], EPM [27], HTCN [28], I^{3}Net [57], and SFA [30]. For the deep learning methods, the models are trained on the labeled source domain dataset and tested on the target domain dataset to ensure a consistent initial setup with other DA methods. These five compared domain adaptation methods are currently the classical approaches with high performance in both the SAR domain and the field of computer vision.

The cross-sensor SAR detection tasks are categorized into four scenarios, that is, FARAD Ka data adaptation to MiniSAR data, MiniSAR data adaptation to FARAD Ka data, FARAD X data adaptation to MiniSAR data, and MiniSAR data adaptation to FARAD X data. For ease of representation, MiniSAR data are denoted by M, FARAD Ka data are denoted by Ka, and FARAD X data are denoted by X. Taking FARAD Ka data adaptation to MiniSAR data as an example, the training data consist of labeled FARAD Ka data and unlabeled MiniSAR data, with the test data consisting of MiniSAR data. This task is denoted by Ka→M. Similarly, we derived the tasks Ka→M, X→M, M→Ka, and M→X. The generalization performance of the detection models is evaluated by using four metrics, PR, RE, $F1$, and mAP. The detection results for these four tasks are shown in Tables III–VI, and the visualization results are depicted in Figs. 10–13.

In these four tasks, the mAP performance of the deep learning methods is all less than 45%, showing poor generalization performance in cross-sensor detection performance. Among these deep learning methods, RTMDet shows better performance, which can reach a mAP of 41.69% on Ka→M task. However, the mAP of RTMDet is still about 30% lower than that of our proposed methods. Therefore, the deep learning methods cannot be applied to cross-sensor tasks directly. In the Ka→M cross-sensor detection task, as shown in Table III and visualized in Fig. 10, our method achieved the mAP of 68.71%, which is about 7% higher than the second-ranked EPM algorithm and more than 10% higher

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TABLE III
DETECTION RESULTS (%) ON KA→M TASK. THE BEST AND SECOND BEST VALUES ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY

Methods	Ka → M			
	PR	RE	F1	mAP
Faster-RCNN [42]	47.83	43.26	45.43	33.06
DINO [54]	56.50	39.04	46.18	40.05
YOLOF [55]	43.46	46.63	44.99	36.87
RTMDet [56]	41.13	53.37	46.45	41.69
DAF [26]	64.53	59.27	61.79	54.24
EPM [27]	72.52	53.37	61.49	61.28
HTCN [28]	57.92	59.55	58.73	53.15
I^3Net [57]	55.18	50.84	52.92	46.37
SFA [30]	58.10	53.37	55.64	54.32
Our method	**76.45** (+3.93)	**62.92** (+3.37)	**69.03** (+7.24)	**68.71** (+7.43)

TABLE V
DETECTION RESULTS (%) ON M→KA TASK. THE BEST AND SECOND BEST VALUES ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY

Methods	M → Ka			
	PR	RE	F1	mAP
Faster-RCNN [42]	35.59	35.56	35.58	21.54
DINO [54]	35.29	36.71	35.98	26.64
YOLOF [55]	31.06	35.56	33.16	23.22
RTMDet [56]	33.47	41.81	37.18	28.82
DAF [26]	51.87	54.85	53.32	40.57
EPM [27]	57.24	43.27	49.29	45.88
HTCN [28]	49.44	41.71	45.25	32.09
I^3Net [57]	44.94	43.07	43.98	36.38
SFA [30]	42.87	37.02	39.73	28.34
Our method	**68.20** (+10.96)	**57.25** (+2.40)	**62.25** (+8.93)	**55.12** (+9.24)

TABLE IV
DETECTION RESULTS (%) ON X→M TASK. THE BEST AND SECOND BEST VALUES ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY

Methods	X → M			
	PR	RE	F1	mAP
Faster-RCNN [42]	47.67	40.17	43.60	26.53
DINO [54]	48.31	36.24	41.41	34.62
YOLOF [55]	39.05	41.57	40.27	30.23
RTMDet [56]	44.27	39.04	41.49	35.11
DAF [26]	59.33	59.83	59.58	49.91
EPM [27]	72.39	54.49	62.18	61.02
HTCN [28]	62.07	50.56	55.73	44.44
I^3Net [57]	49.86	50.84	50.35	42.17
SFA [30]	56.59	53.09	54.78	48.11
Our method	**73.58** (+1.19)	**72.75** (+12.92)	**73.16** (+10.98)	**68.51** (+7.49)

TABLE VI
DETECTION RESULTS (%) ON M→X TASK. THE BEST AND SECOND BEST VALUES ARE MARKED IN BOLD AND UNDERLINED, RESPECTIVELY

Methods	M → X			
	PR	RE	F1	mAP
Faster-RCNN [42]	46.82	36.32	40.91	26.42
DINO [54]	45.86	43.06	44.14	35.38
YOLOF [55]	36.72	38.39	37.54	28.14
RTMDet [56]	38.20	46.82	42.07	34.63
DAF [26]	59.59	60.05	59.82	50.72
EPM [27]	66.83	54.60	60.10	53.44
HTCN [28]	47.96	51.88	49.84	44.59
I^3Net [57]	54.15	48.25	51.03	45.05
SFA [30]	45.51	49.29	47.32	39.44
Our method	**69.99** (+3.16)	**63.81** (+3.76)	**66.76** (+6.66)	**62.56** (+9.12)

than other algorithms. In addition, the PR and RE results of the proposed method were 76.45% and 62.92%, respectively. The $F1$ score of the proposed method was 69.03%. These three metrics also reached the best among all algorithms. The visualization results show that the HTCN and DAF methods produced a higher number of false alarms and detected fewer targets. Compared with these methods, our proposed method can highlight the salient features of targets, suppress surrounding background clutter, and significantly reduce false alarms and missed detections.

In the X→M cross-sensor detection task, as shown in Table IV and visualized in Fig. 11, our method achieved mAP and $F1$ scores of 68.51% and 73.16%, respectively. The PR and RE of the proposed method also showed a great advantage over other algorithms. The HTCN algorithm, which uses CycleGAN for image transfer, performed only slightly above 44% in the X→M cross-sensor detection task. The possible reason is that the HTCN algorithm only migrates the style of the entire image, neglecting the alignment of target scattering features and details, leading to bad performance. The SFA algorithm, which combines transformers with domain knowledge in cross-domain transfer, experienced an mAP reduction of nearly 20% compared with our proposed method. This may be because the performance of the transformer network relies on the volume of data. More data can lead to better performance. The limitation of the volume of SAR data and the complexity of SAR image interpretation may lead to poor performance on the mAP metrics. Due to the dense arrangement of targets, many methods such as DAF and EPM produced a significant number of missed detections, with around ten targets missed. Our method, in comparison, only missed two targets, showing a significant performance advantage. The possible reason is that our method can better represent the domain-invariant essential features of targets, focusing on the target perception center for more accurate localization.

Furthermore, in the M→Ka and M→X cross-sensor detection tasks, our method also achieved excellent results, showing more advantages over other methods, as shown in Tables V and VI, and visualized in Figs. 12 and 13. In the M→Ka and M→X tasks, the detection performance of these algorithms was significantly lower than that of the Ka→M and X→M tasks. The possible reason is that the MiniSAR data contains fewer targets than FARAD data, leading to insufficient transfer of features. In addition, FARAD data contain more diverse vehicle targets in urban areas, with more complex postures and scenes, and is more affected by background clutter. In contrast, MiniSAR data contain vehicle targets in more open

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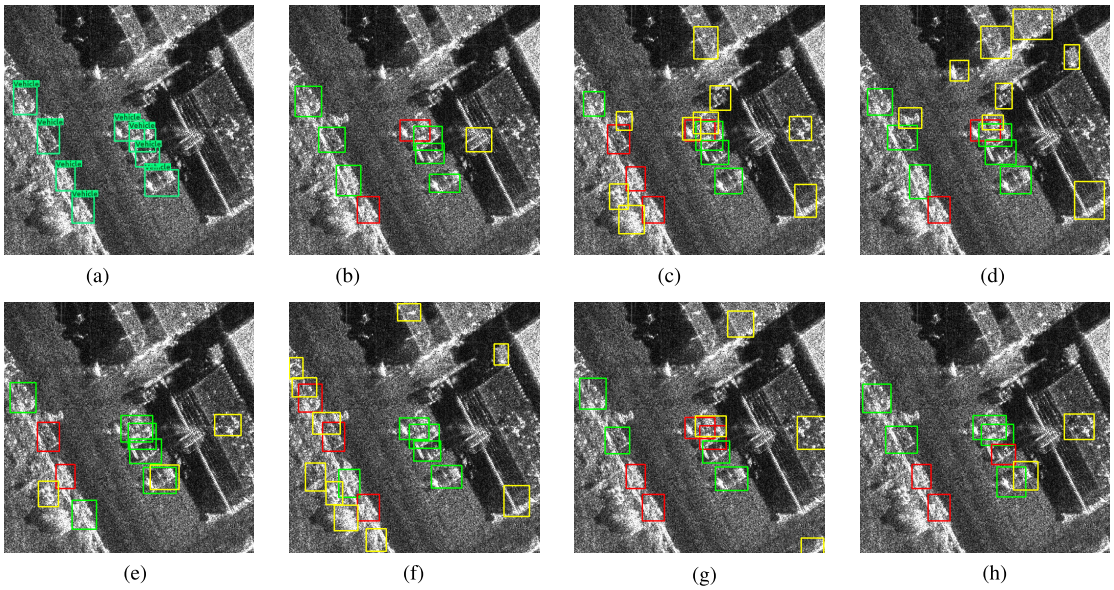


Fig. 10. SAR images are tested on the SOTA methods to compare the performance. The visualization of detection results in the cross-sensor detection task Ka→M. In (a)–(h), green indicates a correctly detected object, red indicates an undetected object, and yellow indicates a wrong detected object. (a) Ground truth. (b) Proposed method. (c) Faster-RCNN. (d) DAF. (e) EPM. (f) HTCN. (g) I^3Net. (h) SFA.

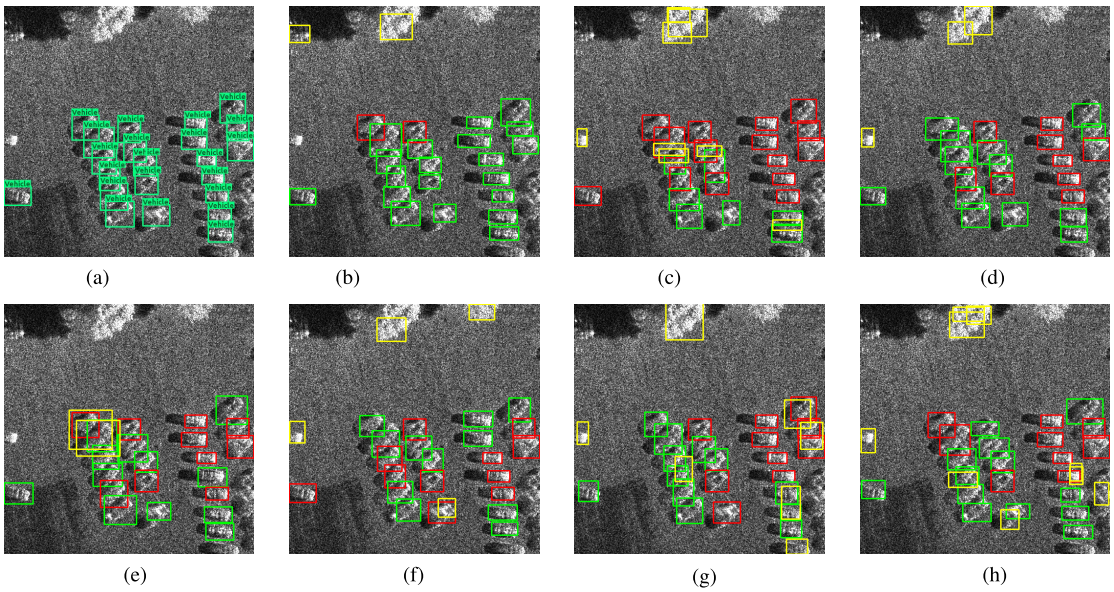


Fig. 11. SAR images are tested on the SOTA methods to compare the performance. The visualization of detection results in the cross-sensor detection task X → M. In (a)–(h), green indicates a correctly detected object, red indicates an undetected object, and yellow indicates a wrong detected object. (a) Ground truth. (b) Proposed method. (c) Faster-RCNN. (d) DAF. (e) EPM. (f) HTCN. (g) I^3Net. (h) SFA.

countryside areas, less affected by buildings and trees. Therefore, the test results for MiniSAR data are better than those for FARAD data. As also shown in Figs. 12 and 13, our proposed method can greatly suppress the impact of background clutter and false alarms, avoiding the misalignment and redundancy caused by negative samples. Especially in the M→X detection task, as shown in Fig. 13, our method detected all targets without false alarms, validating the effectiveness of the approach.

From the results, the cross-sensor detection mAP in Ka→M and M→Ka were 68.71% and 55.12%, respectively. The cross-sensor detection mAP for X→M and M→X were 68.51% and 62.56%, respectively. The results indicate that the feature adaptation between FARAD X data and MiniSAR data is more effective. This indirectly suggests that FARAD X data and MiniSAR data are more similar in terms of target features, image styles, and feature characteristics, allowing for more sufficient feature transfer.

As shown in Figs. 14 and 15, which display the centerness maps and the original images of two tasks, our method pays more attention to the SAR target areas in the images. The proposed method can enhance the salience of targets, which can highlight strong scattering centers separate targets from background negative sample areas, thus improving target detection performance. Moreover, by focusing more on target areas, our method can extract domain-invariant features from different domains, enhancing the accuracy of feature

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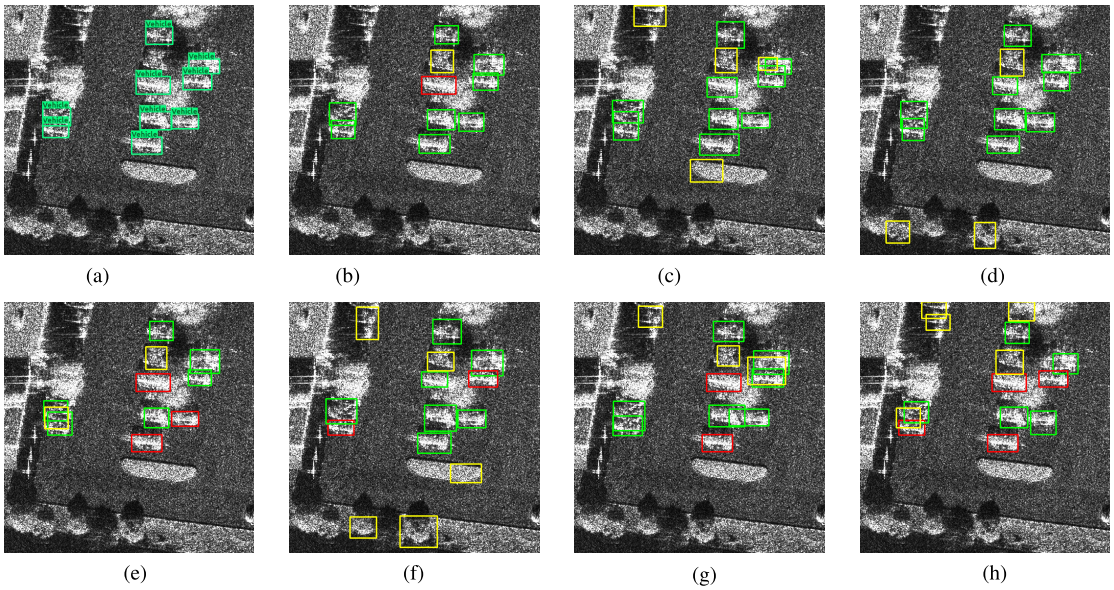


Fig. 12. SAR images are tested on the SOTA methods to compare the performance. The visualization of detection results in the cross-sensor detection task M → Ka. In (a)–(h), green indicates a correctly detected object, red indicates an undetected object, and yellow indicates a wrong detected object. (a) Ground truth. (b) Proposed method. (c) Faster-RCNN. (d) DAF. (e) EPM. (f) HTCN. (g) I^3Net. (h) SFA.

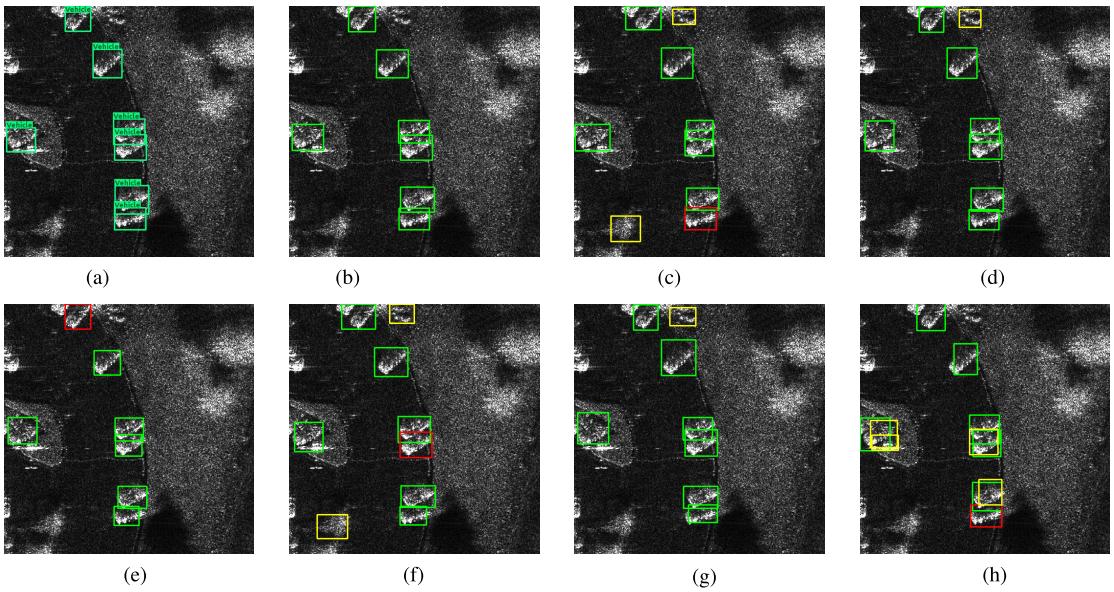


Fig. 13. SAR images are tested on the SOTA methods to compare the performance. The visualization of detection results in the cross-sensor detection task M → X. In (a)–(h), green indicates a correctly detected object, red indicates an undetected object, and yellow indicates a wrong detected object. (a) Ground truth. (b) Proposed method. (c) Faster-RCNN. (d) DAF. (e) EPM. (f) HTCN. (g) I^3Net. (h) SFA.

calibration and the generalization capability of cross-sensor detection. Overall, our proposed method achieves better performance compared with other algorithms, validating the effectiveness of the proposed method.

In addition, the centerness maps of MiniSAR generated in two tasks are visualized as shown in Fig. 16. Compared with the Ka→M task, the targets in centerness map obtained in the X→M task are more prominent with clearer surrounding target area and more salient target scattering centers. In addition, the separability between the target and the background is better, which can lead to more accurate localization of the target. Since other imaging parameters have less differences among these sensors, we analyze the reasons mainly from the perspective of band characteristics. The X band and Ku band have a closer range of frequencies. The images of these two bands have similar image styles, gray intensity, and feature details. The target features are more accurately characterized when feature transfer is performed in our proposed method, leading to better domain adaptation.

Due to the similarity between MiniSAR and FARAD X data, it can further enhance the center-aware calibration of scattering centers and component structures during the domain adaptation process, which can lead to more accurate localization of target positions and more distinction between targets. Therefore, better performance can be obtained on the proposed algorithm when the experiments are performed on

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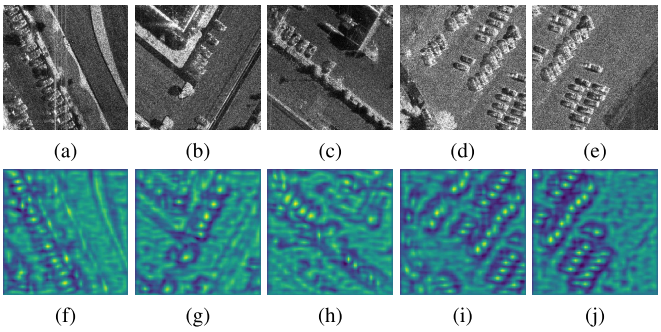


Fig. 14. Visualization of the centerness map of MiniSAR data on Ka→M. (a)–(e) Five different original images of MiniSAR data. (f)–(j) Corresponding centerness maps of the five original images.

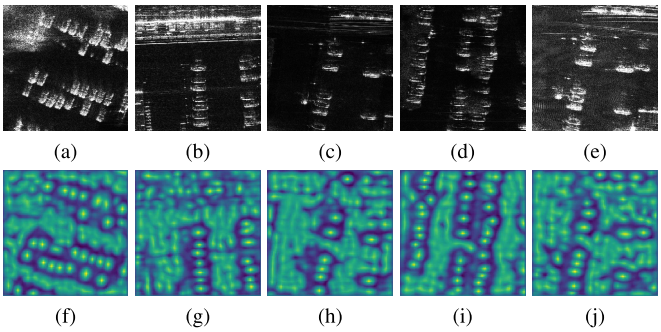


Fig. 15. Visualization of the centerness map of FARAD X data on M→X. (a)–(e) Five different original images of MiniSAR data. (f)–(j) Corresponding centerness maps of the five original images.

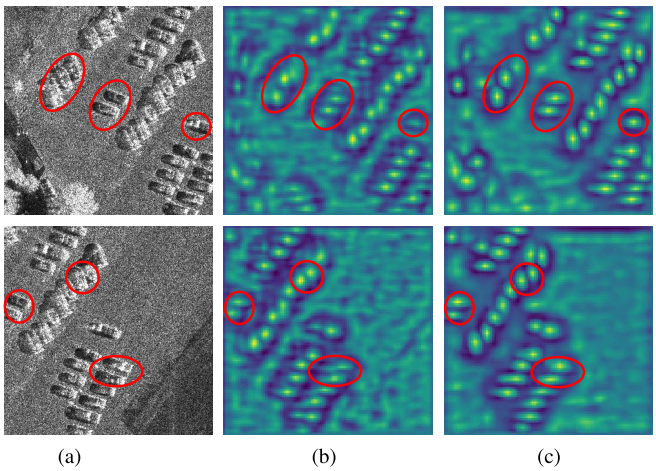


Fig. 16. Comparative visualization of the centerness maps in Ka→M and X→M tasks in the inference process. (a) Original images. (b) Ka→M. (c) X→M.

the domain adaptation detection tasks of X→M and M→X. However, compared with the X band, the Ka band has a higher range of frequency and is susceptible to the inference of background clutter. The high similarity between the target and the background makes it susceptible to negative samples during dynamic feature alignment. This can result in high false alarms in the Ka→M and M→Ka tasks.

D. Ablation Experiments

To compare the robustness of the algorithms and the detection performance of different modules, the ablation experiments are carried out and the performance of the algorithms is analyzed in terms of the backbone network, different modules, the number of feature-aligned layers, and different loss weights.

TABLE VII
RESULTS OF MAP (%) WITH DIFFERENT BACKBONES

Backbone	Ka → M	M → Ka	X → M	M → X
MobileNetv2 [58]	56.96	36.64	41.97	37.14
VggNet-16 [59]	61.86	40.06	47.49	38.01
VggNet-19 [59]	55.79	35.10	54.84	37.89
ResNet-34 [60]	63.17	35.91	59.43	36.38
ResNet-50 [60]	**68.71**	**55.12**	**68.51**	**62.56**
ResNet-101 [60]	67.85	49.64	65.96	53.70

1) Impact of Backbone: The selection of feature extraction backbone networks can significantly influence the performance of cross-sensor SAR target detection. In this section, we analyze the mAP performance of six classical backbones on cross-sensor target detection tasks, including MobileNetv2 [58], VggNet-16 [59], VggNet-19 [59], ResNet-34 [60], ResNet-50 [60], and ResNet-101 [60], as shown in Table VII.

Our experimental results indicate that ResNet-50 can achieve the best performance among the six backbone networks in the cross-sensor tasks in this article. Among the above backbone networks, the MobileNetv2 has the comparative worst performance. Compared with the ResNet architectures, VggNet lacks a residual network structure, resulting in inferior feature extraction capabilities and weaker generalization for target detection. Similarly, compared with ResNet-50, ResNet-34 has fewer residual blocks. Therefore, its feature extraction ability is worse and the feature representation is not that accurate. Conversely, while ResNet-101 boasts a more complex structure and enhanced feature extraction capabilities compared with ResNet-50, its detection performance is relatively poor. Notably, in cross-sensor tasks of M→Ka and M→X, ResNet-101 underperforms ResNet-50 by approximately 5% in terms of the result of mAP. This discrepancy may stem from the complexity and deeper layers of ResNet-101, which makes it converge more readily on the dataset of the source domain. Furthermore, given the limited volume of SAR data, overfitting becomes a concern, particularly within deeper network architectures, hindering the generalization of features to new datasets and thus adversely affecting cross-sensor detection performance. ResNet-50, which has similar network structure modules to ResNet-101 and residual connections, exhibits robust feature extraction capabilities. Moreover, its fewer layers facilitate the generalization of features to new datasets. Consequently, ResNet-50 is selected as the backbone network for feature extraction in the cross-sensor target detection algorithm in this article.

2) Impact of Different Modules: The contribution of different modules to the algorithm varies; hence; ablation studies were designed to evaluate the impact of each module. The method without the participation of either module is denoted as "source only," which signifies training and testing without domain adaptation techniques, relying solely on the baseline. The method without the CACM is labeled as "w/o CACM,"

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TABLE VIII
MAP RESULTS(%) ON ABLATION EXPERIMENTS OF DIFFERENT MODULES ON FARAD KA AND MINISAR ADAPTATION TASKS

Method	Ka → M				M → Ka			
	PR	RE	F1	mAP	PR	RE	F1	mAP
Source only	45.42	39.05	41.99	34.16	39.24	21.48	27.76	17.85
w/o CACM	67.70	61.24	64.31	63.72	62.63	48.59	54.73	45.39
w/o DFDM	75.10	55.06	63.53	63.57	60.87	52.56	56.41	50.46
Our method	**76.45**	**62.92**	**69.03**	**68.71**	**68.20**	**57.25**	**62.25**	**55.12**

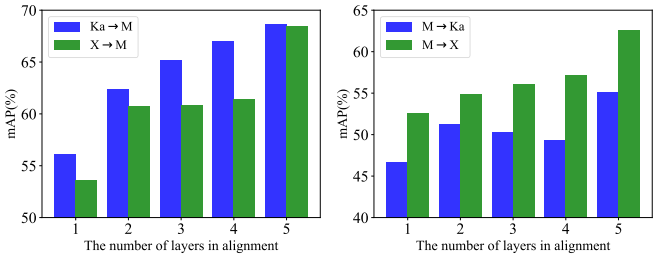
TABLE IX
MAP RESULTS(%) ON ABLATION EXPERIMENTS OF DIFFERENT MODULES ON FARAD X AND MINISAR ADAPTATION TASKS

Method	X → M				M → X			
	PR	RE	F1	mAP	PR	RE	F1	mAP
Source only	46.32	30.06	36.46	27.90	41.03	37.35	39.10	26.28
w/o CACM	71.04	57.87	63.78	62.13	64.89	63.04	63.95	57.57
w/o DFDM	65.25	64.89	65.07	63.47	66.77	54.22	59.84	57.99
Our method	**73.58**	**72.75**	**73.16**	**68.51**	**69.99**	**63.81**	**66.76**	**62.56**

and the method without the DFDM is labeled as "w/o DFDM." The detection results are presented in Tables VIII and IX.

The "source only" method yields the poorest detection performance, with an mAP not exceeding 35% across four cross-sensor detection tasks, particularly with an mAP of only 17.85% in the M→Ka task. When employing the method "w/o DFDM," where only the CACM module is involved in training, there is a noticeable improvement in detection performance. In the Ka→M and X→M tasks, the mAP values increase to 63.57% and 63.47%, respectively. In the M→Ka and M→X tasks, the mAP reaches 50.46% and 57.99%, respectively. In the "w/o CACM" method, where only the DFDM module is involved in training, detection performance also improves. The mAP values in the Ka→M and X→M tasks reach 63.72% and 62.13%, respectively. In the M→Ka and M→X tasks, the mAP values are 45.39% and 57.57%, respectively. In the proposed method, when both modules participate in training, the mAP values in the Ka→M and X→M tasks exceed 68%. This indicates that both modules contribute positively to the method and the combined involvement in detection tasks significantly enhances performance. The same conclusion can be drawn for the M→Ka and M→X tasks. In the four cross-sensor SAR target detection tasks, the proposed method improves by more than 5% compared with training with only one module. However, analysis of the ablation study on the four cross-sensor detection tasks reveals that when trained individually, the DFDM generally outperforms the CACM, especially in the M→Ka and X→M tasks. The "w/o DFDM" method outperforms the "w/o CACM" method by about 5% in terms of mAP on M→Ka task. This suggests that the DFDM plays a more significant role in these two tasks. When both modules are involved in training simultaneously, detection performance is significantly enhanced.

3) Impact of the Number of Layers Involved in Alignment: The method proposed in this article employs a multiscale feature alignment strategy, utilizing multiscale structures within two modules to enhance the representation of features during discrimination and calibration. This can integrate high-level and low-level features, mitigating the loss of scattered information that can result from increased network dimensions. The feature alignment method, which combines various scales, further enhances the diversity of feature alignment within the algorithm, improving the localization performance of detection and accuracy rate. To further validate the robustness of the algorithm, we analyze the impact of the number of feature alignment layers on algorithm performance. In the ablation study, the number of feature alignment layers varied from 1 to 5, examining the influence of the number of alignment layers on the results. To maintain the consistency of the feature alignment, the number of layers involved in DFDM and CACM are kept the same with each other. The comparison was conducted on four cross-sensor SAR target detection tasks, with the results illustrated in the bar charts shown in Fig. 17.



(a) (b)

Fig. 17. (a) and (b) Bar charts of the mAP under different number of layers in alignment. (a) Based on the Ka→M and X→M tasks. (b) Based on the M→Ka and M→X tasks.

It can be observed that as the number of alignment layers increases, the detection performance generally trends upward. In the X→M task, as depicted in Fig. 17(a), when the number of layers is less than 5, the mAP values are all below 65%. In the M→Ka task, as depicted in Fig. 17(b), when the number of layers is 5, the mAP value exceeds 55%, but when the number of layers is less than 5, the mAP is mostly below 50%. The reason for this is that in lower dimensional network layers, the features extracted are texture details and structure characteristics of the target, whereas, in higher dimensional network layers, the features extracted are complex and abstract semantic features. By combining features of different dimensions, the alignment and discrimination of features become more accurate and comprehensive, leading to better cross-sensor target detection performance and generalization capabilities.

4) Impact of the Weight of the Losses: The algorithm loss proposed in this article mainly contains two weights λ and μ. To further analyze the impact of different combinations of weights on the performance of the algorithm, experiments are conducted using different combinations of weights for four cross-sensor tasks. The results are shown in Fig. 18.

Initially, λ is set to 0.5, and μ/λ takes the values of 0, 0.5, 1.0, 1.5, and 2, respectively, and the test results are obtained, as shown in Fig. 18(a) and (b). From the analysis of the experimental results, the maximum value of mAP can be obtained when the μ/λ ratio is 1. When μ/λ takes the values

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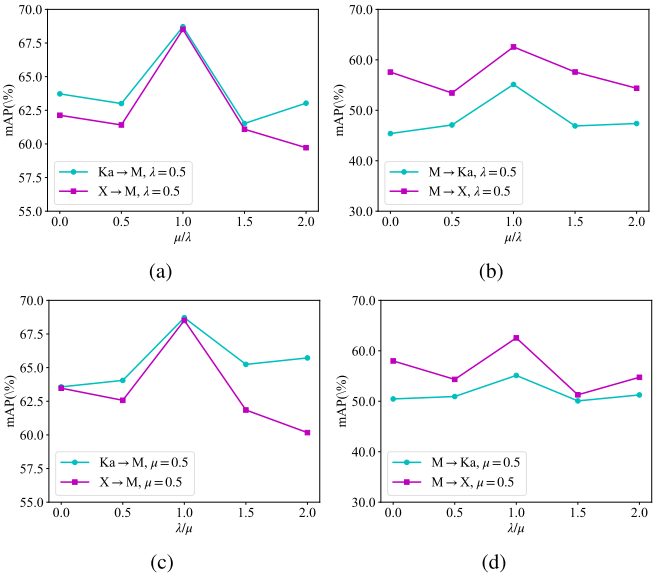


Fig. 18. (a)–(d) Line charts of the mAP under different combination of the loss weights. (a) and (c) Based on the Ka→M and X→M tasks. (b) and (d) Based on the M→Ka and M→X tasks.

of 0.5 and 1.5, mAP decreases by about 5%-10%. Then, μ is set to 0.5, and the values of λ/μ are taken as 0, 0.5, 1.0, 1.5, and 2, and the detection results are obtained as shown in Fig. 18(c) and (d). Similarly, when the ratio of λ/μ is 1, the mAP can reach the maximum value. In the X→M task, when the ratio is 2, the mAP result is less than 60%. When ratios are 0.5, the mAP is less than 65% in both Ka→M and X→M tasks. It is analyzed that when the ratio of λ and μ is too large or too small, the two modules cannot be fully utilized. When λ is larger than μ, less attention is paid to the target perception center alignment, resulting in lower performance. However, when λ is smaller than μ, less attention is paid to the dynamic feature alignment of the image, resulting in lower performance. Therefore, when λ and μ are equal and are both set to 0.5, the better performance can be achieved.

V. Conclusion

To address the challenge of poor generalization and adaptability in SAR target detection when training and testing data are derived from different sensors, this article introduces a cross-sensor SAR target detection method based on dynamic feature discrimination and center-aware calibration. On one hand, a DFDM has been designed to address the escalation of the offset in the bounding box regression. In this module, a BSFA mechanism and a multiscale structure are combined to aggregate features in both horizontal and vertical directions and to integrate high-level and low-level features. This can precisely constrain target locations, compensate for the loss of scattering information, and enhance the discrimination capability. On the other hand, a CACM has been developed to address deviation of the feature alignment. In this module, the target salience relationship is modeled and the center of the target is focused by combining the centerness map and classification map. This can suppress background clutter interference and calibrate the domain-invariant features, which alleviates the misalignment caused by negative samples within bounding boxes. Experimental results demonstrate the effectiveness of our approach. In the future, we will continue to explore cross-sensor SAR target detection and investigate generalization algorithms for SAR target detection under open conditions.

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